

A top-down view of a light teal desk. In the top left is a silver laptop with a black keyboard. To its right is a white mouse and a dark coffee cup. In the bottom left is a brown leather notebook with a white pen and a pair of tortoiseshell glasses.

AIClub

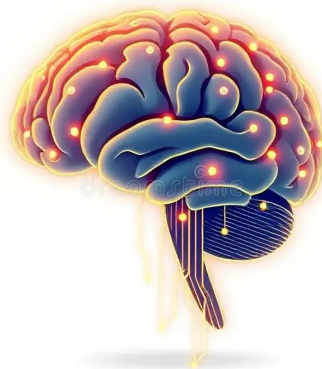




Introduction to Artificial Intelligence and Machine Learning

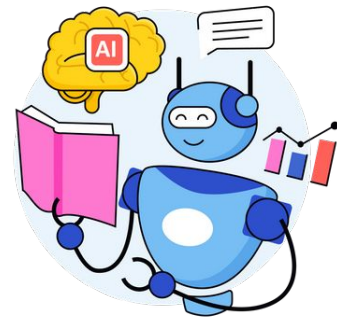
What is Intelligence?

- Ability to learn
- Ability to reason
- Ability to solve problems
- Ability to adapt



What is Artificial Intelligence (AI)?

Artificial Intelligence is the ability of machines to perform tasks that normally require human intelligence.

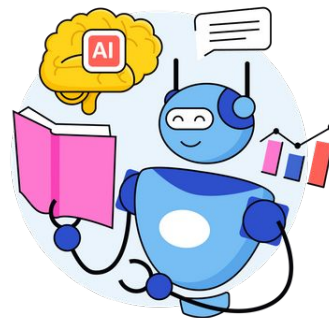


What is Artificial Intelligence (AI)?

Artificial Intelligence is the ability of machines to perform tasks that normally require human intelligence.

Examples:

- Face unlock
- Google Maps
- Chatbots
- Recommendation systems

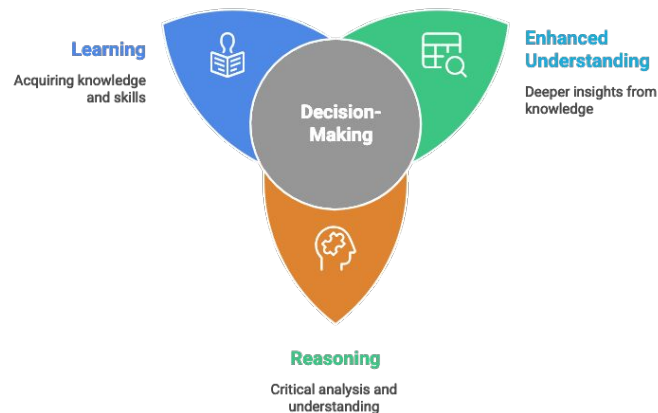


Formal Definition of AI

Artificial Intelligence is a branch of computer science that enables machines to mimic human intelligence such as learning, reasoning, and decision-making.

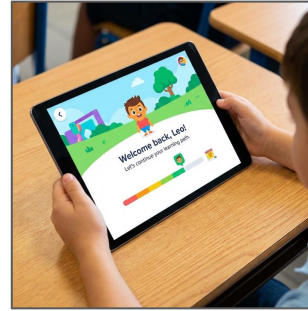
Key Words Highlighted:

- ✓ Learning
- ✓ Reasoning
- ✓ Decision-making



Real-Life Applications of AI

- Education (Personalized learning apps)
- Healthcare (Disease prediction)
- Transportation (Self-driving cars)
- Entertainment (Netflix/YouTube recommendations)

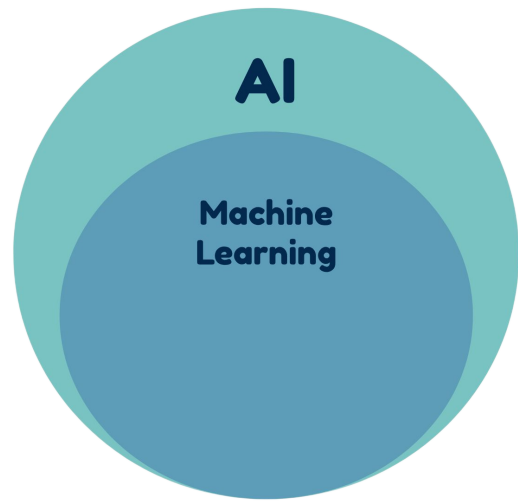


What is Machine Learning (ML)?

Machine Learning is a subset of AI that allows machines to learn from data and improve without being explicitly programmed.

Important Point:

- AI is a big concept
- ML is one way to achieve AI



Types of Machine Learning

Supervised

Provide a lot of examples so that they understand the concept for future reference



Unsupervised

No information is given but by observing, the AI would learn on its own



Reinforcement

A reward system for teaching the AI



AI vs Machine Learning

Artificial Intelligence

- Broad concept
- Makes machines intelligent
- Can be rule-based

Machine Learning

- Subset of AI
- Helps machines learn from data
- Always data-driven

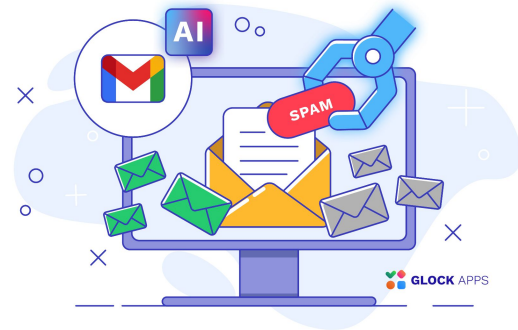
Example to Understand ML

Example: Spam Email Filter

- Instead of manually writing rules, the system learns from thousands of spam emails.
- It improves over time.

Working flow:

- Data → Pattern → Prediction



Quick Classroom Activity

Think–Pair–Share

Identify whether the example is:

- AI
- ML
- Both

Think



Pair



Share



Quick Classroom Activity

Think–Pair–Share

Identify whether the example is:

- AI
- ML
- Both
- None

Examples:

- Calculator
- Face Recognition
- Auto-correct
- Netflix Recommendation
- Chess-playing computer

A top-down view of a desk with a light blue background. In the top left, a silver laptop is partially visible. Next to it is a white mouse. In the bottom left, there is a dark brown leather notebook with a white pen and a pair of tortoiseshell glasses.

THANK YOU



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<https://aiclub.world>